Aryavart Dahiya

Final Year Undergraduate
Department of Mechanical Engineering
Indian Institute of Technology Kanpur

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Examination	Degree/Certificate	Institute	Year	CPI/%
Graduation	B.Tech	Indian Institute of Technology Kanpur	2023–Present	**8.2**/10
Intermediate/+2	CBSE(XII)	Sky Hawk International School, Delhi	2023	**92.40**%
Matriculation	CBSE(X)	Model School, Rohtak	2021	**96.20**%

Work Experiences

Deutsche Bank (Received PPO) *May'26 – Jul'26*
Software Developer Intern

- Owned **end-to-end** development of **CSTM**, a **full-stack** platform built with **Spring Boot, Java** and **React**, replacing a legacy mapping system and driving through **production deployment**
- Gathered and translated requirements directly from **Business Functional Analysts (BFAs)** into technical specifications, driving feature scope, **UI/UX** design, and the prioritization decisions
- Architected and integrated **DBLLM**, an internal **LLM-powered** service, using **Java WebClient**, enabling one-click generation of **PySpark** & **JavaSpark** transformation code from population rules
- Optimized the **LLM** inference pipeline through structured **input processing**, **prompt** engineering and **data standardization**, significantly reducing hallucinations and cutting end-to-end code generation latency from **1–1.5 minutes** to under **20 seconds**
- Designed and implemented scalable **feed cloning**, **selective mapping** replication and **bulk ingestion** workflows, enabling efficient, reliable management of **thousands** of mapping feeds
- Engineered scalable **REST APIs** and backend services supporting **100–200+ mappings** per feed, with production-ready validation and data workflows, adopted across multiple **business teams**

DataPKT Aviation *Dec'24 – Feb'25*
Data Engineer Intern

- Engineered scalable **Python–SQL data pipelines** to process **10M+** drone telemetry records, enabling efficient real-time analytics and faster data retrieval for mission-critical operations
- Developed an interpretable **XGBoost model** for predictive maintenance across **5+** critical aviation subsystems, leveraging domain-specific feature engineering to achieve **95.2%** accuracy
- Improved operational efficiency by reducing unplanned system downtime by **30%** and **accelerating analytics** workflows by **25%** through **predictive forecasting** and optimized **data processing**

Major Competition

Governance Control Plane for AI Agents *Jul'26 – Aug'26*
National Finalist, American Express CodeStreet 2026

- Built a fail-closed **AI agent** policy engine in **Python** with **FastAPI REST/WebSocket APIs**, enforcing least-privilege permissions, **separation-of-duties**, and spend/velocity limits at **0.7 ms p50**
- Engineered **concurrency-safe SQLite-WAL state** with atomic cap decrements, sustaining ∼**1,700** authorization decisions/s with zero double-spend under sustained heavy parallel agent load
- Implemented **Ed25519** request-signing for agent identity, with replay protection and key rotation, preventing identity spoofing
- Built **tamper-evident** audit logging and kill-switches, defending against **5** rogue-agent threat classes; validated with **149 tests**

Scholastic Achievements

- Declared **National Finalists (Top 6)** in American Express Hackathon (Codestreet) 2026 among **15000+** teams nationwide
- Achieved **Codeforces Expert** rating of **1665**, ranking highly among the worldwide competitive programming community
- Ranked **771** and **684** globally (of **30,000+** total participants) in Codeforces Rounds 1031 and 1033 (Division 2) respectively
- Awarded **KVPY Extended Fellowship** by successfully securing **All India Rank 1638** in the KVPY exam among **1.5 L+** candidates

Position Of Responsibility

Coordinator, English Literary Society *Apr'25 – Apr'26*

- Led a **29-member executive** team, managing a **INR 1L+ budget** to organize institute-wide literary events and workshops

Key Projects

Centralized Gymkhana Facilities Platform *Jan'26 – Apr'26*
Software Dev & Ops - Course Project *Prof. Indranil Saha*

- Led a **9-member team** to develop and deploy a centralized Gymkhana facilities platform using **Node.js, Express.js, MongoDB** & **React.js**, architecting a scalable full-stack system
- Implemented secure **role-based access control (RBAC)**, **OAuth 2.0**, and **JWT**-based authentication, while carefully designing robust and normalized database schemas and **REST APIs** for bookings, user management, and various facility operations
- Built **core modules** including real-time slot booking, **QR-based authentication**, and **AI-powered** facial verification, and automated end-to-end deployments seamlessly through **GitHub Actions CI/CD** and fully production-grade release pipelines
- Authored comprehensive **unit** and **integration** test suites across frontend and backend, thoroughly covering **route-level API** testing and multi-role workflows for 6+ distinct user roles

Distributed In-Memory Key-Value Store *Nov'25 – Jan'26*
Self Project

- Built a **Redis-inspired** distributed key-value store in **C++**, supporting **1M+** key-value pairs with **multithreaded TCP** networking, configurable persistence & sub-millisecond lookups
- Designed lock-efficient concurrent **hash tables**, **LRU eviction**, append-only persistence and asynchronous replication using **thread pools**, reducing latency under concurrent workloads
- Implemented a custom **Redis-compatible** command protocol, connection pooling, and benchmark suite, achieving **100K+ operations/sec** while maintaining consistency across replicas

Memory-Efficient File Indexing Engine *Feb'26 – Mar'26*
CS253 Course Project *Prof. Indranil Saha*

- Built a memory-efficient **C++ CLI** file indexer with a **3-layer** architecture, efficiently designed for **110+ MB** versioned files
- Implemented generic templates, runtime **polymorphism**, and configurable **chunked I/O** for scalable, efficient file indexing
- Optimized **Word Count, Delta Diff, and Top-K** query analytics using efficient **hash indices** and **O(N log K)** partial sorting

Companion Arbiter PUF *Aug'25 – Nov'25*
Machine Learning - Course Project *Prof. Purushottam Kar*

- Mathematically demonstrated vulnerability of a **Physically Unclonable Function** to decryption, using a linear classifier
- Enhanced accuracy to **99.3%** and reduced the training time to **1.99s** by engineering **528 features** from the original 32 features
- Examined the efficacy of **LinearSVC** and **LogisticRegression** by optimizing hyperparameters like tolerance, loss and penalty

Facial Lesion Inpainting *Jan'25 – Apr'25*
CV & Deep Learning - Course Project *Prof. K. Jerripothula*

- Developed an automated facial lesion restoration system using **Telea** inpainting, **DIP-based U-Net**, and precise **lesion detection**
- Automated **71-point** facial landmark masking using **Dlib** and **FAN**, achieving **20%+** better localization and **0.498 recall**

Technical Skills

- **Languages & Tools:** C | C++ | Java | Python | SQL | HTML | CSS | Bash | Git | Docker | OAuth 2.0 | JWT | REST APIs | WebSockets
- **Libraries & Frameworks:** Spring Boot | React.js | Next.js | FastAPI | Node.js | PySpark | PostgreSQL | OpenCV | Scikit-learn

Relevant Courses

† Online * Outstanding

Software Dev & Operations*
Fundamentals of Computing
Computer Vision & Deep Learning
Data Structures & Algorithms†
Intro to DBMS†
Machine Learning
Probability & Statistics†
Signal, Systems & Networks*